Phase-dependent recovery and lifetime of encoded entanglement under collective detector noise

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Abstract

We study how a local preparation phase controls the recovery and lifetime of encoded reference entanglement when two qubits of a four-qubit register experience collective detector noise. The preparation family has identical initial reference–register negativity and identical detector marginals. Within a secular weak-coupling generator, an exact trace-norm reduction proves a phase optimum at every time. A specified Dicke-basis measurement and conditional phase correction recover a two-qubit resource whose negativity equals a phase factor times the total variation distance between two evolved sector populations. Weak singlet–triplet mixing produces a metastable lifetime, obtained by projection onto the slow sector. Applying this resource law to established accelerated-vacuum and inertial thermal spectra gives the leading lifetime ratio at matched local rates and an observation time maximizing the resource contrast. Common-coupling protection is exact under the stated symmetry; the finite-separation lifetime has a conditional microscopic interpretation through ordered weak-coupling and small-separation limits. Optimality is restricted to the preparation family, and recovery refers to the specified measurement protocol.

Keywords: Encoded entanglement, Collective noise, Quantum recovery, Metastability, Unruh–DeWitt detectors

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1. Introduction

The usefulness of an entangled register depends on how its correlations are encoded and recovered, as well as on the noise acting on its physical constituents. Quantum error correction, noiseless subsystems and decoherence-free subspaces provide established ways to exploit this distinction [1, 2, 3]. For a concrete encoding, a useful question is whether a restricted preparation operation can improve the entanglement retained by a specified recovery procedure, without changing the initial reference entanglement or the marginal state exposed to the environment.

We examine this question for a four-qubit cluster-state encoding [4]. One qubit is a reference, one is an uncoupled register qubit, and two are detectors interacting with a common field. Recent work reported maximal reference–register negativity when only one detector of a four-qubit cluster state is coupled to the field [5]. This single-detector protection follows from erasure correctability. When both detectors interact, the encoding must instead be compared with the singlet–triplet structure of the joint noise. The marginal detector state alone does not determine this compatibility.

Local rotations have already been used to compare the robustness of equivalent cluster-state representations under independent dephasing [6]. Superoperator studies also compare four-qubit cluster entanglement with the accuracy of measurement-based operations under independent local noise [7]. Thus basis-dependent robustness and the distinction between entanglement and task performance are established. Here only two detectors are noisy, the preparation phase preserves their joint marginal, and the noise has collective transition channels. The question is the exact phase dependence of this encoded resource and its specified recovery, including the slow loss of protection when sectors mix.

Our main result is an exact expression for the reference–register negativity under a collective secular weak-coupling generator. It proves an all-time phase optimum within a one-parameter family of local preparations. A specified detector measurement and conditional phase correction recover a two-qubit resource whose negativity equals a phase factor times the classical distinguishability of two sector populations. A small violation of collective coupling turns the symmetry-protected resource into a metastable one; a two-dimensional projection determines its slow decay rate and phase-dependent amplitude. These results concern a stated preparation family and recovery procedure, rather than optimization over all encodings and recoveries.

Accelerated detectors supply an application with a controlled comparison of environments. The local response in accelerated vacuum has the Unruh temperature [8], while two-position correlations differ from those in an inertial thermal bath [9]. We use the established detector spectra and population dynamics of Ref. [9]; their derivation is not a new result here. The additional question is how these spatial correlations affect the lifetime of encoded reference entanglement and the recovered resource. Matching the local rates yields a leading lifetime ratio and an analytic observation time maximizing the chosen resource contrast.

Metastability is understood as separation of fast and slow relaxation scales [10]. We distinguish exact protection enforced by common-coupling symmetry from a finite-separation lifetime calculated with an effective master equation. Spatial regulation and an ordered weak-coupling then small-separation limit specify the conditional microscopic interpretation of that lifetime [11, 12, 13, 14].

Section 2 defines the encoding and permitted recovery. Section 3 states the collective dynamics. Section 4 proves the phase optimum and derives the readout resource. Section 5 obtains its slow lifetime and the environmental comparison. Sections 6 and 7 specify the approximation and conclusions. Supplementary Material Sections S1–S5 provide the encoding identities, field spectra, further trace-norm details, slow projection and stability analysis; Section S6 records the numerical checks.

2. Encoding and allowed recovery

Let A be the reference and BCD the register; only C, D couple to the environment. Joint recovery operations within BCD are allowed after the interaction. Spectator free phases are absorbed into the interaction picture and recovery gates. Set Planck's constant, the speed of light and Boltzmann's constant to one, $\hbar = c = k_B = 1$, and use the normalized negativity

$$\mathcal{N}(\rho) = \|\rho^{T_A}\|_1 - 1, \qquad \|Q\|_1 = \mathrm{tr}\sqrt{Q^\dagger Q}. \tag{1}$$

Here T_A is partial transpose on A, tr is the trace, and $Q^\dagger$ denotes the adjoint of an operator Q. This is twice the convention of Ref. [15]; its maximum across $A|BCD$ is one. It measures the specified bipartition, not genuine four-party entanglement.

Apply a Pauli flip X_D and $R_C(\theta) = \mathrm{diag}(1, e^{i\theta})$ to $|\mathrm{CL}_4\rangle = (|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle)/2$. The resulting family is

$$|\psi_\theta\rangle = \frac{|00\rangle_{AB}(|01\rangle + e^{i\theta}|10\rangle)_{CD} + |11\rangle_{AB}(|01\rangle - e^{i\theta}|10\rangle)_{CD}}{2}. \tag{2}$$

Every member has $\mathcal{N}(0) = 1$ and the same detector marginal,

$$\rho_{CD}(0) = \tfrac{1}{2}(P_t + P_s), \quad |t\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad |s\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \quad P_j = |j\rangle\langle j|. \tag{3}$$

Set $|g\rangle = |00\rangle$, $|e\rangle = |11\rangle$ and $P_T = P_g + P_t + P_e = I - P_s$, where I is the identity on CD.

The original cluster encoding satisfies the erasure form of the Knill–Laflamme condition for loss of either C or D [1]. For example, discarding D leaves an equal mixture of $P^{AB}_{\Phi^+} \otimes P^C_0$ and $P^{AB}_{\Phi^-} \otimes P^C_1$, where $|\Phi^\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$. The orthogonal C flags retain negativity one. A trace-preserving channel on D leaves this reduced state unchanged, so monotonicity and the qubit-reference upper bound force the full negativity to remain one. Detector exchange and local preparation unitaries establish the other cases. This explains single-detector protection without a relativistic assumption; Supplementary Material S1 gives the code conditions.

For joint noise, consider a channel $\mathcal{E}$ admitting Kraus operators such that

$$\mathcal{E}(\rho) = \sum_\mu K_\mu \rho K_\mu^\dagger, \qquad [K_\mu, P_s] = 0, \qquad \sum_\mu K_\mu^\dagger K_\mu = I. \tag{4}$$

This sector-preservation condition is stronger than exchange covariance: it forbids transfer between singlet and triplet. Denote the identity channel on AB by id_{AB}.

Proposition 1 (Sector protection)**.** *For a channel satisfying Eq. (4),*

$$\mathcal{N}\Big[(\mathrm{id}_{AB} \otimes \mathcal{E})(|\psi_\theta\rangle\langle\psi_\theta|)\Big] \geq |\sin\theta|. \tag{5}$$

Proof. Resolve $|\psi_\theta\rangle$ as $(|\phi_T\rangle|t\rangle + |\phi_s\rangle|s\rangle)/\sqrt{2}$, where

$$|\phi_{T,s}\rangle = \tfrac{1}{2}[(1 \pm e^{i\theta})|00\rangle + (1 \mp e^{i\theta})|11\rangle].$$

Both normalized AB states have negativity $|\sin\theta|$. After the channel, a non-selective measurement of P_T, P_s leaves these states with orthogonal detector flags, each of weight $1/2$. The trace norm adds over the two flag blocks, giving negativity $|\sin\theta|$. A register-side measurement cannot increase negativity, proving the bound. □

At $\theta = \pi/2$, a fixed register unitary recovers a Bell pair:

$$V = \mathrm{diag}(1, i)_B \otimes P_T + \mathrm{diag}(1, -i)_B \otimes P_s, \tag{6}$$

$$(I_A \otimes V)(\mathrm{id}_{AB} \otimes \mathcal{E})(P_{\psi_{\pi/2}})(I_A \otimes V^\dagger) = P_{\Phi^+} \otimes \mathcal{E}(P_\xi), \tag{7}$$

$$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2},$$

$$|\xi\rangle = (e^{i\pi/4}|t\rangle + e^{-i\pi/4}|s\rangle)/\sqrt{2}.$$

To verify Eq. (7), the two controlled gates map $|\phi_T\rangle, |\phi_s\rangle$ at $\theta = \pi/2$ to $e^{i\pi/4}|\Phi^+\rangle$ and $e^{-i\pi/4}|\Phi^+\rangle$, respectively. Thus V factors the initial state into the Bell pair and $|\xi\rangle$; commuting V through each $I_B \otimes K_\mu$ gives the output identity. The phase selects maximally entangled conditional AB states even when inter-sector coherence is lost.

The decoder could also be applied before the noise, transferring the Bell pair to the uncoupled AB subsystem. Equation (7) identifies compatibility of the encoding with sector-preserving noise. It does not establish an advantage over arbitrary joint preprocessing, or a protocol restricted to local operations among four separated parties.

3. Collective detector dynamics

Take identical two-level detectors of energy gap $\omega > 0$, with downward and upward rates $\gamma_\downarrow, \gamma_\uparrow$. For the detector density matrix ρ, the secular weak-coupling generator is [9, 16, 17]

$$\dot\rho = -i[H_{\mathrm{eff}}, \rho] + \sum_{\eta=\pm1} (1 + \eta f)\Big(\gamma_\downarrow \mathcal{D}[J_\eta]\rho + \gamma_\uparrow \mathcal{D}[J_\eta^\dagger]\rho\Big), \tag{8}$$

$$J_\pm = (\sigma_C^- \pm \sigma_D^-)/\sqrt{2}, \qquad \mathcal{D}[J]\rho = J\rho J^\dagger - \tfrac{1}{2}\{J^\dagger J, \rho\}.$$

The dot denotes differentiation with respect to proper time τ, $\sigma_j^- = |0\rangle_j\langle 1|$, and braces denote the anticommutator. The real spatial factor obeys $|f| \le 1$. The renormalized Hamiltonian H_{eff} is diagonal in the Dicke basis (g, t, e, s) [18]; its triplet–singlet energy difference changes a coherence phase without changing the negativity below. Define the rate ratio, local relaxation rate and dimensionless time by

$$q = \gamma_\uparrow/\gamma_\downarrow, \quad \Gamma = \gamma_\downarrow + \gamma_\uparrow, \quad z = \Gamma\tau, \quad r_\pm = \frac{1 \pm f}{1 + q}. \tag{9}$$

We use $0 < q < 1$ for the thermal slow-limit analysis; $q = 0$ is separately accessible exactly. At $f = 1$ only the symmetric ladder is active and Eq. (4) holds. At $|f| < 1$ both ladders are active.

The population column vector $\boldsymbol{p} = (p_g, p_t, p_e, p_s)^{\mathsf{T}}$ uses ordinary transpose T. Its evolution is $d\boldsymbol{p}/dz = M\boldsymbol{p}$, with the established rate matrix

$$M = \begin{pmatrix} -q(r_+ + r_-) & r_+ & 0 & r_- \\ qr_+ & -(1+q)r_+ & r_+ & 0 \\ 0 & qr_+ & -(r_+ + r_-) & qr_- \\ qr_- & 0 & r_- & -(1+q)r_- \end{pmatrix}. \tag{10}$$

For example, the triplet population has gain $qr_+p_g + r_+p_e$ and loss $(1+q)r_+p_t$; the other rows follow from the same two jump ladders. The columns sum to zero. With $\mathcal{E}_z$ the detector channel at z, define $p_j^{(t)} = \langle j|\mathcal{E}_z(P_t)|j\rangle$, $p_j^{(s)} = \langle j|\mathcal{E}_z(P_s)|j\rangle$, and $d_j = p_j^{(t)} - p_j^{(s)}$. Their difference evolves as

$$\boldsymbol{d}(z) = e^{Mz}(0, 1, 0, -1)^{\mathsf{T}}. \tag{11}$$

At $|f| < 1$ and $q > 0$, the unique stationary state is the product Gibbs state and $\boldsymbol{d}$ vanishes at long times. At $f = 1$, the stationary singlet and the relaxed triplet remain distinguishable.

4. Exact phase optimum and a specified readout

Let $\mathcal{E}_z$ denote the channel generated by Eq. (8), and define $|\zeta_\pm\rangle = (|01\rangle \pm e^{i\theta}|10\rangle)/\sqrt{2}$, $W_\theta(z) = \mathcal{E}_z(|\zeta_+\rangle\langle\zeta_-|)$, $h = |\sin\theta|$ and $E = e^{-2z}$.

Proposition 2 (Exact negativity and phase optimum)**.** *For the encoding in Eq. (2) and the generator in Eq. (8),*

$$\mathcal{N}_\theta(z) = \|W_\theta(z)\|_1, \tag{12}$$

$$\mathcal{N}_\theta(z) = \frac{h}{2}(|d_g| + |d_e|) + \sqrt{E + h^2 K(z)}, \tag{13}$$

$$K(z) = \tfrac{1}{4}\Big[d_t^2 + d_s^2 - 2E + 2|d_t d_s + E|\Big] \geq 0. \tag{14}$$

Consequently $e^{-z} = \mathcal{N}_0(z) \leq \mathcal{N}_\theta(z) \leq \mathcal{N}_{\pi/2}(z)$ at every finite time. The maximizing phase is $\pi/2$ modulo π, with possible degeneracy.

Proof. Write $|\psi_\theta\rangle = (|00\rangle|\zeta_+\rangle + |11\rangle|\zeta_-\rangle)/\sqrt{2}$. Partial transpose on A puts the cross terms into an independent $AB = 01, 10$ block,

$$\tfrac{1}{2}\begin{pmatrix} 0 & W_\theta^\dagger \\ W_\theta & 0 \end{pmatrix}.$$

Its eigenvalues are plus and minus half the singular values of W_θ. The remaining two blocks are positive and each has trace $1/2$. This proves Eq. (12).

Put $\alpha = \frac{1}{2}\sin\theta$, $b = (1+\cos\theta)/2$ and $c = (1-\cos\theta)/2$, so $bc = \alpha^2$. In the Dicke basis,

$$|\zeta_+\rangle\langle\zeta_-| = i\alpha(P_t - P_s) + b|t\rangle\langle s| + c|s\rangle\langle t|. \tag{15}$$

The diagonal part evolves to $i\alpha\sum_j d_j P_j$. For $|t\rangle\langle s|$, all jump-gain terms vanish, while the upward and downward anticommutators sum to $-\Gamma|t\rangle\langle s|$. Thus $\mathcal{E}_z(|t\rangle\langle s|) = e^{-z-i\varphi}|t\rangle\langle s|$, where $\varphi = (E_t - E_s)z/\Gamma$ and E_t, E_s are the corresponding Hamiltonian eigenenergies. The scalar g, e entries of W_θ contribute $|\alpha|(|d_g| + |d_e|)$; the remaining block is

$$Y = \begin{pmatrix} i\alpha d_t & be^{-z-i\varphi} \\ ce^{-z+i\varphi} & i\alpha d_s \end{pmatrix}. \tag{16}$$

For its two singular values, the sum of squares is $\mathrm{tr}(YY^\dagger)$ and the product is $|\det Y|$. Therefore $\|Y\|_1^2 = \mathrm{tr}(YY^\dagger) + 2|\det Y|$, with

$$\mathrm{tr}(YY^\dagger) = \alpha^2(d_t^2 + d_s^2) + (b^2 + c^2)E, \qquad \det Y = -\alpha^2(d_t d_s + E).$$

Using $b^2 + c^2 = 1 - h^2/2$ proves Eqs. (13)–(14); the Hamiltonian phase cancels.

If $d_t d_s + E \geq 0$, then $K = (d_t + d_s)^2/4$. Otherwise $(d_t - d_s)^2 \geq -4d_t d_s > 4E$ gives $K = [(d_t - d_s)^2 - 4E]/4 \geq 0$. The right-hand side of Eq. (13) is nondecreasing in $h \in [0, 1]$, which proves the optimum and its lower bound. □

At $f = 0$, Eq. (10) gives $d_g = d_e = 0$, $d_t = -d_s = e^{-z}$; every phase then has $\mathcal{N} = e^{-z}$. At finite z, the positive term E in Eq. (13) ensures $\mathcal{N}_\theta > 0$. These statements concern $A|BCD$; they do not imply the absence of sudden death for another partition or entanglement measure.

An operational readout measures CD in the Dicke basis and applies an outcome-dependent phase correction to B, with no postselection. It produces

$$\mathcal{N}_{\mathrm{rec},\theta}(z) = \frac{h}{2}\sum_{j=g,t,e,s}|d_j(z)| = h\,\mathrm{TV}(\boldsymbol{p}^{(t)}, \boldsymbol{p}^{(s)}) \leq \mathcal{N}_\theta(z), \tag{17}$$

$$\rho_{AB}^{\mathrm{rec}} = \tfrac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|) + \tfrac{\mathcal{N}_{\mathrm{rec},\theta}}{2}(|00\rangle\langle 11| + |11\rangle\langle 00|). \tag{18}$$

For outcome j, the unnormalized conditional AB coherence is $\langle 00|\rho_{AB}^{(j)}|11\rangle = \langle j|W_\theta|j\rangle/2 = i\alpha d_j/2$. A phase gate on B, chosen from the known preparation and dynamics, makes each entry real and nonnegative. Summing all outcomes yields the off-diagonal entry $h\sum_j |d_j|/4$, while the two diagonal entries sum to $1/2$ each. Partial transpose of Eq. (18) then gives Eq. (17). Thus no outcome is discarded.

Here $\mathrm{TV}(\boldsymbol{p},\boldsymbol{q}) = \frac{1}{2}\sum_j |p_j - q_j|$ is total variation distance. The subscript "rec" refers to this specified procedure; it is not a maximization over all recovery maps. At $z = 0$ it yields h, although the full register has negativity one. The standard Bell-measurement teleportation protocol using Eq. (18) has the known average-fidelity relation [19]

$$\overline{F} = \frac{2 + \mathcal{N}_{\mathrm{rec},\theta}}{3}. \tag{19}$$

Indeed, Eq. (18) is a mixture of $\Phi^{\pm}$ with weights $(1 \pm \mathcal{N}_{\mathrm{rec},\theta})/2$; standard teleportation gives a dephasing channel with transverse Bloch factors $\mathcal{N}_{\mathrm{rec},\theta}$. Averaging a pure input over the Bloch sphere gives Eq. (19). This connects resource decay to a familiar task; no new teleportation protocol is claimed.

The collective channel resolves symmetric and antisymmetric amplitudes, so a relative detector phase changes the encoded sector correlations even though the detector marginal is unchanged. At $f = 0$, independent phase-covariant detector noise removes this dependence, as the exact solution above confirms. The decay mechanism is loss of the distinguishable sector flag. Even if the initial t–s coherence decays, the conditional AB entanglement survives while the two evolved sector populations remain distinguishable. Sector mixing reduces their total variation distance and hence the specified recovered resource.

At zero temperature, $q = 0$, the exact population solution gives the useful boundary check

$$\mathcal{N}_\theta = e^{-z}\left[h|\sinh(fz)| + \sqrt{1 + h^2\sinh^2(fz)}\right], \qquad \mathcal{N}_{\pi/2} = e^{-(1-|f|)z}. \tag{20}$$

Supplementary Material S3 gives the population solution; this boundary result does not require fixed-$q > 0$ perturbation theory.

5. Metastable lifetime and an environmental comparison

5.1. Weak sector mixing

Set $\epsilon = 1 - f \ll 1$, $Z = 1 + q + q^2$ and

$$\mathcal{C}(q) = 1 + \frac{q}{Z} = \frac{(1+q)^2}{Z}. \tag{21}$$

At $\epsilon = 0$ the stationary vectors are $\boldsymbol{r}_T = (1, q, q^2, 0)^{\mathsf{T}}/Z$ and $\boldsymbol{r}_s = (0, 0, 0, 1)^{\mathsf{T}}$.

Proposition 3 (Slow resource law)**.** *At fixed $0 < q < 1$, the weakly mixed generator has a physical slow decay rate*

$$\kappa = \epsilon\Gamma\mathcal{C}(q) + O(\epsilon^2\Gamma). \tag{22}$$

At fixed slow time $u = \epsilon z > 0$,

$$\begin{aligned} \boldsymbol{d}(u/\epsilon) &\longrightarrow e^{-\mathcal{C}(q)u}(\boldsymbol{r}_T - \boldsymbol{r}_s), \\ \mathcal{N}_\theta(u/\epsilon),\ \mathcal{N}_{\mathrm{rec},\theta}(u/\epsilon) &\longrightarrow h\, e^{-\mathcal{C}(q)u}. \end{aligned} \tag{23}$$

Proof. Write $M = M_0 + \epsilon M_1$, where $M_0 = M|_{f=1}$ and M_1 follows by differentiating Eq. (10) with respect to ϵ. The left zero vectors $\boldsymbol{\ell}_T = (1, 1, 1, 0)$ and $\boldsymbol{\ell}_s = (0, 0, 0, 1)$ obey $\boldsymbol{\ell}_i \boldsymbol{r}_j = \delta_{ij}$. Projection gives

$$[\boldsymbol{\ell}_i M_1 \boldsymbol{r}_j]_{i,j=T,s} = \begin{pmatrix} -q/Z & 1 \\ q/Z & -1 \end{pmatrix}. \tag{24}$$

For example, the singlet-row entries are q/Z and -1; column conservation fixes the other row. The two eigenvalues are zero and $-\mathcal{C}(q)$. The other eigenvalues of M_0 are $-2 \pm 2\sqrt{q}/(1+q)$ and remain separated from zero. Finite-dimensional perturbation about this isolated two-dimensional subspace gives one stationary eigenvalue and one eigenvalue $-\epsilon\mathcal{C}(q) + O(\epsilon^2)$; multiplication by Γ gives Eq. (22).

The initial difference has sector coordinates $(1, -1)^{\mathsf{T}}$ and zero total mass. Its slow projection is $\boldsymbol{r}_T - \boldsymbol{r}_s$, the difference eigenvector. At $z = u/\epsilon$, the fast population modes and the coherence e^{-z} vanish, while the difference mode tends to $e^{-\mathcal{C}(q)u}$. Substitution into Eqs. (13) and (17) proves Eq. (23), since the two limiting population vectors have disjoint supports and $\|\boldsymbol{r}_T - \boldsymbol{r}_s\|_1 = 2$. Errors are $O(\epsilon)$ on compact positive intervals of u, with an additional exponentially small fast contribution. Supplementary Material S4 gives M_1 explicitly. □

Thus the specified readout attains the full negativity to leading order on the slow scale. For a general phase the fast initial relaxation from one to h is a boundary layer; the limit is not uniform down to $u = 0$. Even at the optimal phase, finite separation gives initial slope $\mathcal{N}_{\pi/2}(z) = 1-\epsilon z+O(z^2)$ for $f \geq 0$, obtained by expanding $\boldsymbol{d} = \boldsymbol{d}(0) + Mz\boldsymbol{d}(0) + O(z^2)$ in Eq. (13). The metastable plateau is not exact freezing. Figure 1 shows the collective phase dependence and the convergence on the slow scale. The factor $\mathcal{C}(q) = 1+q/Z$ is the sum of the singlet-to-triplet rate and the reverse rate after rapid triplet equilibration.

5.2. Matched local rates in accelerated vacuum and a thermal bath

Consider identical detectors uniformly accelerated with acceleration $a > 0$ and separated transversely by L, interacting with a massless scalar field in four-dimensional Minkowski vacuum through a dimensionless weak coupling λ. Compare inertial detectors at the same separation in a thermal bath of inverse temperature $2\pi/a$. Rates are λ^2 times the field spectra at $\pm\omega$. The point-profile spectra, derived in Supplementary Material S2 following Ref. [9], give

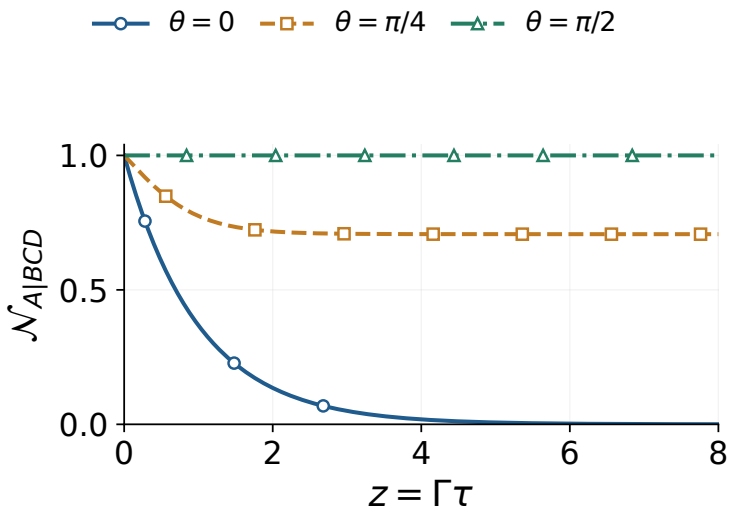
$$q = e^{-2\pi\omega/a}, \qquad \gamma_\downarrow = \frac{\lambda^2\omega}{2\pi(1-q)}, \qquad \gamma_\uparrow = q\gamma_\downarrow, \tag{25}$$

$$f_{\rm acc} = \frac{\sin[(2\omega/a)\,\mathrm{arsinh}(aL/2)]}{\omega L\sqrt{1+a^2L^2/4}}, \qquad f_{\rm th} = \mathrm{sinc}(\omega L). \tag{26}$$

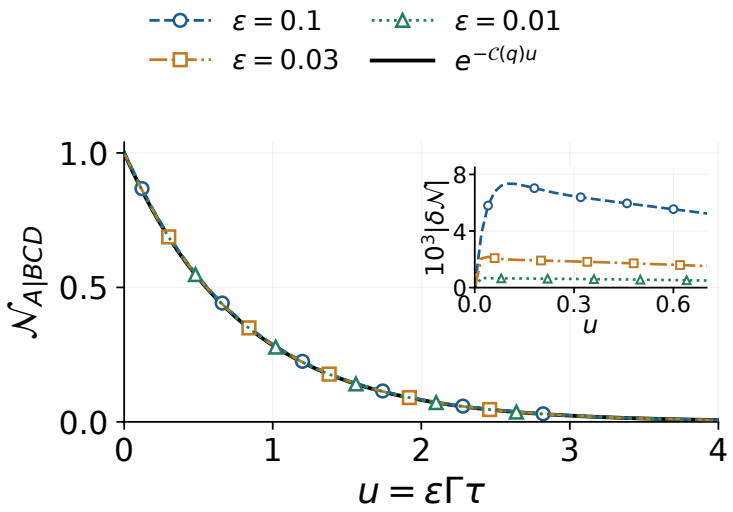
Here $\mathrm{sinc}\,x = \sin x/x$ with $\mathrm{sinc}\,0 = 1$, and arsinh is the inverse hyperbolic sine. The local rates agree, but the spatial factors do not. The quadratic small-separation expansion requires both $\omega L \ll 1$ and $aL \ll 1$. Expanding the known factors at fixed a, ω gives

$$\epsilon_{\rm acc} = \frac{\omega^2+a^2}{6}L^2 + O(L^4), \qquad \epsilon_{\rm th} = \frac{\omega^2}{6}L^2 + O(L^4),$$
$$\frac{\kappa_{\rm acc}}{\kappa_{\rm th}} \longrightarrow R = 1 + \frac{a^2}{\omega^2}. \tag{27}$$

The accelerated resource therefore has the shorter leading lifetime at matched local rates. The spatial spectrum is an input from Ref. [9]; Eq. (27) translates it into the encoded-resource lifetime.



(a) Exact collective coupling.



(b) Weak sector mixing.

Figure 1: Phase dependence and slow decay at $q = 0.4$. (a) At $f = 1$, the optimal phase retains maximal $A|BCD$ negativity, whereas other phases relax towards $|\sin\theta|$. (b) At $\theta = \pi/2$ and $f = 1 - \epsilon$, the exact generator approaches $e^{-\mathcal{C}(q)u}$ on the slow scale $u = \epsilon\Gamma\tau$. The inset shows the actual absolute errors multiplied by 10^3; curves and markers use unshifted data.

At the optimal phase and fixed $w = \Gamma\omega^2 L^2\tau/6 > 0$ as $L \to 0$, Eq. (23) yields $\mathcal{N}_{\rm th} \to e^{-\mathcal{C}w}$ and $\mathcal{N}_{\rm acc} \to e^{-\mathcal{C}Rw}$, with the same limits for the recovered resource. Maximizing their difference gives, for $R > 1$,

$$w_* = \frac{\ln R}{\mathcal{C}(q)(R-1)}, \qquad \tau_* = \frac{6w_*}{\Gamma\omega^2 L^2}, \tag{28}$$

$$\Delta\mathcal{N}_{\rm max} = (1 - R^{-1})R^{-1/(R-1)}. \tag{29}$$

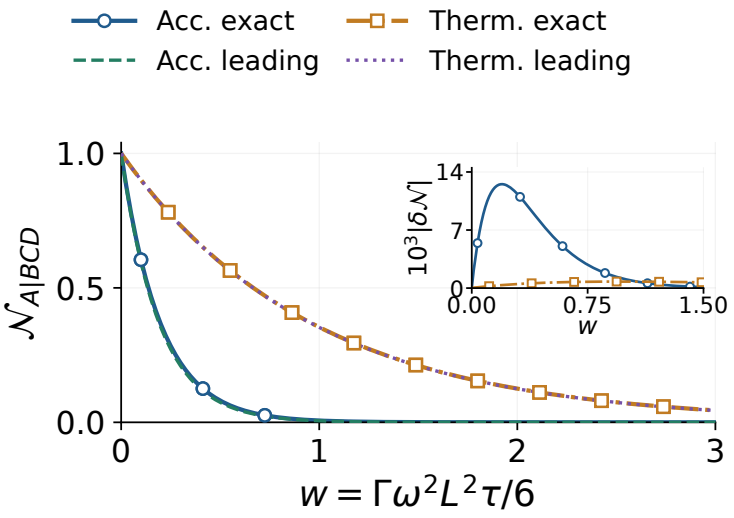
For the derivation, $\Delta(w) = e^{-\mathcal{C}w} - e^{-\mathcal{C}Rw}$ has derivative $\mathcal{C}[-e^{-\mathcal{C}w} + Re^{-\mathcal{C}Rw}]$. Its unique positive zero obeys $e^{\mathcal{C}(R-1)w} = R$. Since Δ starts at zero, is initially increasing and returns to zero at infinity, this point is the global maximum. The contrast tends continuously to zero as $a/\omega \to 0$ and $R \to 1$. Acceleration changes the weak sector-mixing rate through the spatial correlation, rather than through the matched local relaxation rate.

These are leading small-separation formulas. The corresponding teleportation-fidelity contrast is $\Delta\mathcal{N}_{\rm max}/3$. This maximizes the stated resource contrast, not the success probability of an optimized environmental discrimination experiment. Figure 2 compares exact-generator dynamics with the environmental limits; sampled values and absolute errors are supplied with the code.

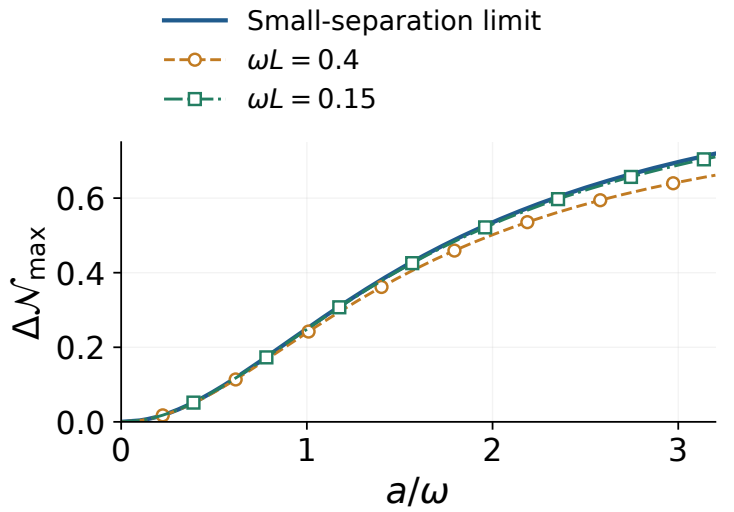
Numerical checks compare the complete four-qubit evolution with the analytic reduction in 800 cases (maximum absolute error 2.13×10^{-15}) and in 300 additional boundary and late-time cases up to $z = 500$ (maximum 2.60×10^{-13}). For $q \in \{0.01, 0.1, 0.4, 0.9\}$, ϵ from 0.1 to 0.002, $u \in \{0.5, 1, 2\}$ and $\theta \in \{0.37, \pi/2\}$, endpoint errors decrease towards the slow law. At $\epsilon = 0.002$, the largest sampled relative slow-rate error is 1.26×10^{-4} and the negativity error is 1.52×10^{-4}. Supplementary Material S6 and the accompanying tables state the grids and contrast tests. These checks support the effective-model calculation; they do not extend its microscopic range.

6. Microscopic scope and sensitivity

Use smooth, identical detector profiles of radius $r < a^{-1}$, stationary under the common boost and translated transversely by L [12]. The register and field are initially independent; detector gaps, couplings and switching functions are identical. At $L = 0$ the smeared field operators coincide. Both the free detector Hamiltonian and the interaction commute with P_s, so the full detector–field unitary does too. Bath matrix elements then give Eq. (4).



(a) Matched local response.



(b) Optimized negativity contrast.

Figure 2: Environmental application at $\theta = \pi/2$. (a) Exact-generator curves and leading slow laws for $a/\omega = 2$, $\omega L = 0.2$, with common $q = e^{-\pi}$ and local rate Γ. The inset gives absolute errors multiplied by 10^3. (b) Maximum contrast from the full generator at the indicated separations, compared with Eq. (29). "Exact" refers throughout to the effective generator, not to the microscopic detector–field evolution.

The protection and decoder in Eqs. (5)–(7) are consequently exact for this common coupling, without a Markov approximation.

At $L > 0$, the lifetime is conditional on the secular weak-coupling description. Conditions include $\Gamma \ll \omega$, $\Gamma\tau_B \ll 1$ for bath correlation time τ_B, and compatible renormalized Hamiltonian scales [13, 14]. These conditions alone do not establish microscopic accuracy uniformly over a lifetime diverging as L^{-2}. For fixed $r, L > 0$, assume weak-coupling convergence uniformly on bounded intervals of $z = \Gamma_{\lambda,r}\tau$. Taking $\lambda \to 0$ first and then $L \to 0$ gives the conditional ordered limit

$$\lim_{L\to 0^+} \lim_{\lambda\to 0} \mathcal{N}_\theta^{\mathrm{mic}}\left(\frac{u}{\Gamma_{\lambda,r}[1 - f_r(L)]}\right) = he^{-\mathcal{C}(q)u}, \qquad u > 0. \tag{30}$$

Here the regulated rate ratio is held fixed. Reflection symmetry and a finite fourth spectral moment give $1 - f_r(L) = c_r L^2 + O(L^4)$. The point-profile coefficients in Eq. (27) require convergence of the corresponding on-shell second moments as $r \to 0$. A simultaneous $L = L(\lambda)$ limit or a fixed-coupling microscopic memory bound is not established.

Let $\mathcal{L}$ be Eq. (8) and $\widetilde{\mathcal{L}}$ a perturbed time-independent generator, both producing completely positive trace-preserving channels. The diamond norm is the trace-norm map bound allowing an arbitrary reference system. If $\|\widetilde{\mathcal{L}} - \mathcal{L}\|_\diamond \le \delta\Gamma$ with dimensionless $\delta \ge 0$, then

$$|\widetilde{\mathcal{N}}_\theta(\tau) - \mathcal{N}_\theta(\tau)| \le 2\delta\Gamma\tau = \frac{2\delta u}{\epsilon}. \tag{31}$$

Indeed, Duhamel's identity expresses the channel difference as the time integral of two channels sandwiching $\widetilde{\mathcal{L}} - \mathcal{L}$. Channels have diamond norm one, giving a bound $\delta\Gamma\tau$. The qubit-reference inequality $|\mathcal{N}(\widetilde{\rho}) - \mathcal{N}(\rho)| \le 2\|\widetilde{\rho} - \rho\|_1$ supplies the factor two. Supplementary Material S5 gives the full norm argument. A sufficient small-error condition on the slow scale is $\delta \ll \epsilon$. Extra dephasing or, in a near-degenerate generator retaining collective terms, a detector-gap mismatch must therefore be compared with $\epsilon\Gamma$, rather than only with Γ. The bound is sufficient and need not be tight.

7. Discussion

Within a fixed local phase family, the exact trace-norm expression proves an all-time optimal preparation and relates a specified recovered resource to sector-population distinguishability. Weak sector mixing gives both its

leading lifetime and the same slow decay of the full reference–register negativity. The accelerated-vacuum comparison applies these results to known spatial spectra and yields an optimum observation time for resource contrast at matched local rates.

Single-detector protection, singlet symmetry, the collective generator and the fidelity relation have established origins. The contribution is their explicit connection through the phase-dependent trace norm, recovery procedure and slow resource law for this encoding. It does not optimize over general encodings or recovery maps. Common-coupling protection is exact under the stated symmetry; the finite-separation lifetime remains an effective-dynamics prediction with the ordered microscopic interpretation in Eq. (30).

Supplementary material

The accompanying Supplementary Material provides proofs, the regulated model and spectral calculation. The reproducibility archive contains Python sources, figure data and verification results.

Declarations

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Declaration of competing interest

The author declares no competing financial or non-financial interests.

Data availability

No experimental data were used. The numerical verification outputs and figure data are included in the accompanying reproducibility archive. The figures can be regenerated from the supplied Python sources.

Code availability

The archive includes `src/model.py`, `src/validate.py`, the additional `src/validate_submission.py`, pinned requirements and reproduction instructions.

Declaration of generative AI and AI-assisted technologies in the writing process

An OpenAI language-model assistant was used to help restructure and edit the manuscript, check algebraic reductions, compare selected references and run the supplied numerical verification code. The author is responsible for reviewing and approving the final scientific content and references before submission.

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